

## PRACTICAL - 2

### OBJECTIVE:

The objective of this practical is to learn how to install and configure Hadoop in a Single-Node Cluster environment. It also helps in understanding HDFS operations such as storing, managing, retrieving, and monitoring big data efficiently.

**AIM: Hadoop Installation, Configuration and HDFS Operations on Windows/Linux**

### Problem Definition

You have recently joined DataAnalytics Inc. as a Hadoop Administrator.

The organization processes large-scale customer, sales, and marketing data using the Hadoop ecosystem. Due to increasing data volumes, the company has decided to migrate its data storage infrastructure from traditional file systems to Hadoop Distributed File System (HDFS). As a Hadoop Administrator, your responsibility is to install and configure

Hadoop in a Single-Node Cluster environment on Windows/Linux, verify the Hadoop services, manage HDFS storage, and perform data management operations.

Management wants a fully functional Hadoop environment that can store, retrieve, and process big data efficiently.

Your task is to install Hadoop, configure the Hadoop ecosystem, start Hadoop services, and perform HDFS operations using Hadoop commands.

**Task:**

- Hadoop Installation & Configuration
  - Access Hadoop Web UI.
1. Create directories in HDFS.
  2. Upload files to HDFS.
  3. List HDFS files and directories.
  4. Display file contents.
  5. Copy files within HDFS.

6. Move files within HDFS.
7. Rename files.
8. Download files from HDFS.
9. Delete files and directories.
10. Check HDFS report.
11. Verify block information.
12. Check Hadoop cluster health.
13. Monitor Hadoop processes.
14. Check storage utilization.

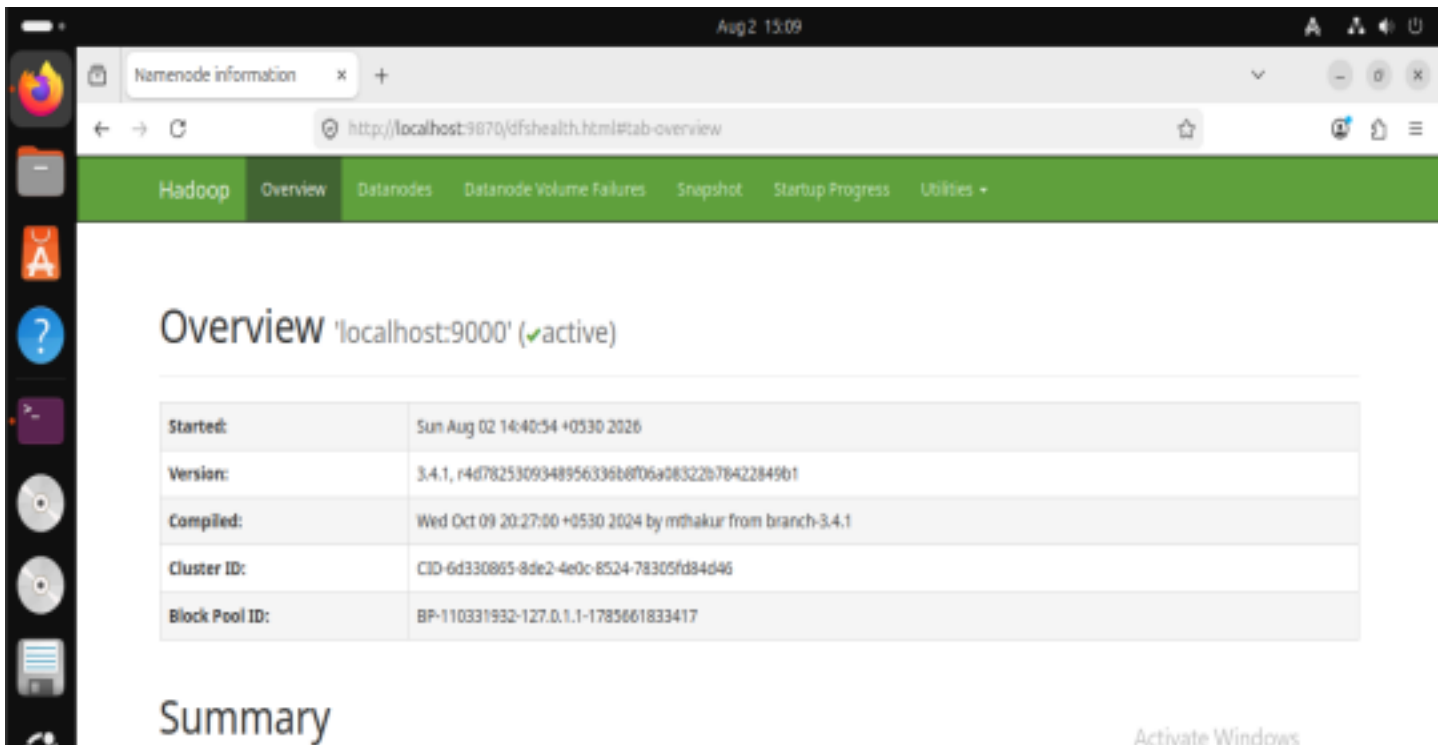
## Tools / Technology Used

- Windows 10 / Windows 11
- OpenJDK 11 or above
- Hadoop 3.x
- Winutils.exe
- Command Prompt / PowerShell

```
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2
~/BDA Practicals/Practical_2

prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/Linux_Practicals/Practical_1
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ cd "/home/prisha-kalola/BDA Practicals"
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals$ pwd
/home/prisha-kalola/BDA Practicals
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals$ mkdir Practical_2
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals$ cd Practical_2
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ pwd
/home/prisha-kalola/BDA Practicals/Practical_2
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ mkdir screenshots
touch sales_data.txt
touch commands.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ ls
commands.txt  sales_data.txt  screenshots
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ nano sales_data.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ cat sales_data.txt
ProductA 120
ProductB 80
ProductC 150
ProductA 280
ProductB 90
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ java -version
openjdk version "11.0.31" 2026-04-21
OpenJDK Runtime Environment (build 11.0.31+11-post-1ubuntu1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 11.0.31+11-post-1ubuntu1-26.04.2-Ubuntu, mixed mode, sharing)
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ readlink -f $(which java)
/usr/lib/jvm/java-11-openjdk-amd64/bin/java
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ hadoop version
Hadoop 3.4.1
```

## Access Hadoop Web UI -



Aug 2 15:09

Namenode information x +

http://localhost:9070/dfshealth.html#tab-overview

Hadoop Overview Datanodes Datanode Volume Failures Snapshot Startup Progress Utilities +

### Overview 'localhost:9000' (✓active)

|                |                                                             |
|----------------|-------------------------------------------------------------|
| Started:       | Sun Aug 02 14:40:54 +0530 2026                              |
| Version:       | 3.4.1, r4d7825309348956336b8f06a08322b78422849b1            |
| Compiled:      | Wed Oct 09 20:27:00 +0530 2024 by mthakur from branch-3.4.1 |
| Cluster ID:    | CID-6d330865-8de2-4e0c-8524-78305fd84d46                    |
| Block Pool ID: | BP-110331932-127.0.1.1-1785661833417                        |

### Summary

Activate Windows

```
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/Linux_Practicals/Practical_1
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practical/Practical_2$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 26.04 LTS
Release:        26.04
Codename:       resolute
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practical/Practical_2$ cd /usr/local/hadoop/etc/hadoop
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:/usr/local/hadoop/etc/hadoop$ ls
capacity-scheduler.xml      kms-log4j.properties
configuration.xml           kms-site.xml
container-executor.cfg      log4j.properties
core-site.xml               mapred-env.cmd
hadoop-env.cmd              mapred-env.sh
hadoop-env.sh               mapred-queues.xml.template
hadoop-metrics2.properties mapred-site.xml
hadoop-policy.xml           shellprofile.d
hadoop-user-functions.sh.example ssl-client.xml.example
hdfs-rbf-site.xml           ssl-server.xml.example
hdfs-site.xml               user_ec_policies.xml.template
httpfs-env.sh               workers
httpfs-log4j.properties    yarn-env.cmd
httpfs-site.xml             yarn-env.sh
kms-acls.xml                yarn-site.xml
kms-env.sh                  yarnservice-log4j.properties
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:/usr/local/hadoop/etc/hadoop$ grep JAVA_HOME hadoop-env.sh
# export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
# Technically, the only required environment variable is JAVA_HOME.
```

```

prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/Linux_Practicals/Practical_1 x prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Pr
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: /usr/local/hadoop/etc/hadoop$ jps
4838 Jps
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: /usr/local/hadoop/etc/hadoop$ sudo nano /usr/local/hadoop/etc/hadoop/core-site.xml
[sudo: authenticate] Password:
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: /usr/local/hadoop/etc/hadoop$ sudo nano /usr/local/hadoop/etc/hadoop/hdfs-site.xml
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: /usr/local/hadoop/etc/hadoop$ mkdir -p ~/hadoopdata/hdfs/namenode
mkdir -p ~/hadoopdata/hdfs/datanode
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: /usr/local/hadoop/etc/hadoop$ ls ~/hadoopdata/hdfs
datanode namenode
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: /usr/local/hadoop/etc/hadoop$ hdfs namenode -format
2026-08-02 14:40:20,343 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG: host = prisha-kalola-VMware-Virtual-Platform/127.0.1.1
STARTUP_MSG: args = [-format]
STARTUP_MSG: version = 3.4.1
STARTUP_MSG: classpath = /usr/local/hadoop/etc/hadoop:/usr/local/hadoop/share/hadoop/common/lib/kerby-pkix-2.0.3.jar:/usr/local/hadoop/sh
/jul-to-slf4j-1.7.36.jar:/usr/local/hadoop/share/hadoop/common/lib/failureaccess-1.0.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-
:/usr/local/hadoop/share/hadoop/common/lib/hadoop-auth-3.4.1.jar:/usr/local/hadoop/share/hadoop/common/lib/jakarta.activation-api-1.2.1.jar
are/hadoop/common/lib/guava-27.0-jre.jar:/usr/local/hadoop/share/hadoop/common/lib/j2objc-annotations-1.1.jar:/usr/local/hadoop/share/hadoo
curity-9.4.53.v20231009.jar:/usr/local/hadoop/share/hadoop/common/lib/gson-2.9.0.jar:/usr/local/hadoop/share/hadoop/common/lib/jackson-core
l/hadoop/share/hadoop/common/lib/kerby-config-2.0.3.jar:/usr/local/hadoop/share/hadoop/common/lib/commons-cli-1.5.0.jar:/usr/local/hadoop/s
b/netty-transport-sctp-4.1.100.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/slf4j-api-1.7.36.jar:/usr/local/hadoop/share/hadoop/comm
ssl-ocsp-4.1.100.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/checker-qual-2.5.2.jar:/usr/local/hadoop/share/hadoop/common/lib/jetty
1009.jar:/usr/local/hadoop/share/hadoop/common/lib/kerb-crypto-2.0.3.jar:/usr/local/hadoop/share/hadoop/common/lib/avro-1.9.2.jar:/usr/loca
/common/lib/jersey-core-1.19.4.jar:/usr/local/hadoop/share/hadoop/common/lib/stax2-api-4.2.1.jar:/usr/local/hadoop/share/hadoop/common/lib/
1.100.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/netty-handler-proxy-4.1.100.Final.jar:/usr/local/hadoop/share/hadoop/common/lib/
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/Linux_Practicals/Practical_1 prisha-kalola@prisha-kalola-VMware-Virtual-Pla
STARTUP_MSG: build = https://github.com/apache/hadoop.git -r 4d7825309340956330b0f06a00322b78422849b1; compiled by 'mthakur
STARTUP_MSG: java = 11.0.31
*****/
2026-08-02 14:40:20,407 INFO namenode.NameNode: registered UNIX signal handlers for [TERM, HUP, INT]
2026-08-02 14:40:20,937 INFO namenode.NameNode: createNameNode [-format]
2026-08-02 14:40:21,539 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-j
2026-08-02 14:40:22,290 INFO namenode.NameNode: Formatting using clusterId: CID-6d330865-8de2-4e0c-8524-78305fd84d46
2026-08-02 14:40:22,388 INFO namenode.FSEditLog: Edit logging is async:true
2026-08-02 14:40:22,460 INFO namenode.FSNamesystem: KeyProvider: null
2026-08-02 14:40:22,464 INFO namenode.FSNamesystem: fsLock is fair: true
2026-08-02 14:40:22,464 INFO namenode.FSNamesystem: Detailed lock hold time metrics enabled: false
2026-08-02 14:40:22,543 INFO namenode.FSNamesystem: fsOwner = prisha-kalola (auth:SIMPLE)
2026-08-02 14:40:22,543 INFO namenode.FSNamesystem: supergroup = supergroup
2026-08-02 14:40:22,544 INFO namenode.FSNamesystem: isPermissionEnabled = true
2026-08-02 14:40:22,544 INFO namenode.FSNamesystem: isStoragePolicyEnabled = true
2026-08-02 14:40:22,545 INFO namenode.FSNamesystem: HA Enabled: false
2026-08-02 14:40:22,891 INFO common.Util: dfs.datanode.fileio.profiling.sampling.percentage set to 0. Disabling file IO profil
2026-08-02 14:40:23,415 INFO blockmanagement.DatanodeManager: Slow peers collection thread shutdown
2026-08-02 14:40:23,470 INFO blockmanagement.DatanodeManager: dfs.block.invalidate.limit : configured=1000, counted=60, effec
2026-08-02 14:40:23,470 INFO blockmanagement.DatanodeManager: dfs.namenode.datanode.registration.ip-hostname-check=true
2026-08-02 14:40:23,470 INFO blockmanagement.BlockManager: dfs.namenode.startup.delay.block.deletion.sec is set to 000:00:00:
2026-08-02 14:40:23,479 INFO blockmanagement.BlockManager: The block deletion will start around 2026 Aug 02 14:40:23
2026-08-02 14:40:23,483 INFO util.GSet: Computing capacity for map BlocksMap
2026-08-02 14:40:23,483 INFO util.GSet: VM type = 64-bit
2026-08-02 14:40:23,486 INFO util.GSet: 2.0% max memory 838 MB = 16.8 MB
2026-08-02 14:40:23,486 INFO util.GSet: capacity = 2^21 = 2097152 entries
2026-08-02 14:40:23,542 INFO blockmanagement.BlockManager: Storage policy satisfier is disabled
2026-08-02 14:40:23,542 INFO blockmanagement.BlockManager: dfs.block.access.token.enable = false

```



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prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/Linux_Practicals/Practical_1
prisha-kalola@prisha-kalola-VMware-V
Starting secondary namenodes [prisha-kalola-VMware-Virtual-Platform]
2026-08-02 14:41:13,687 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using bu
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: /usr/local/hadoop/etc/hadoop$ jps
5828 Jps
5204 NameNode
5513 SecondaryNameNode
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: /usr/local/hadoop/etc/hadoop$ http://localhost:9870
bash: http://localhost:9870: No such file or directory
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: /usr/local/hadoop/etc/hadoop$ cd "/home/prisha-kalola/BDA Practica
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ hdfs dfs -mkdir /sales
2026-08-02 14:43:28,617 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using bu
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ hdfs dfs -put sales_data.txt /sales
2026-08-02 14:43:40,721 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using bu
2026-08-02 14:43:43,943 WARN hdfs.DataStreamer: DataStreamer Exception
org.apache.hadoop.ipc.RemoteException(java.io.IOException): File /sales/sales_data.txt._COPYING_ could only be written
are 0 datanode(s) running and 0 node(s) are excluded in this operation.
    at org.apache.hadoop.hdfs.server.blockmanagement.BlockManager.chooseTarget4NewBlock(BlockManager.java:2473)
    at org.apache.hadoop.hdfs.server.namenode.FSDirWriteFileOp.chooseTargetForNewBlock(FSDirWriteFileOp.java:293)
    at org.apache.hadoop.hdfs.server.namenode.FSNamesystem.getAdditionalBlock(FSNamesystem.java:3875)
    at org.apache.hadoop.hdfs.server.namenode.NameNodeRpcServer.addBlock(NameNodeRpcServer.java:932)
    at org.apache.hadoop.hdfs.protocolPB.ClientNamenodeProtocolServerSideTranslatorPB.addBlock(ClientNamenodeProto
    at org.apache.hadoop.hdfs.protocol.proto.ClientNamenodeProtocolProtos$ClientNamenodeProtocol$2.callBlockingMet
    at org.apache.hadoop.ipc.ProtobufRpcEngine2$Server$ProtoBufRpcInvoker.call(ProtobufRpcEngine2.java:621)
    at org.apache.hadoop.ipc.ProtobufRpcEngine2$Server$ProtoBufRpcInvoker.call(ProtobufRpcEngine2.java:589)
    at org.apache.hadoop.ipc.ProtobufRpcEngine2$Server$ProtoBufRpcInvoker.call(ProtobufRpcEngine2.java:573)
    at org.apache.hadoop.ipc.RPC$Server.call(RPC.java:1227)
    at org.apache.hadoop.ipc.Server$RpcCall.run(Server.java:1246)
    at org.apache.hadoop.ipc.Server$RpcCall.run(Server.java:1169)
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ hdfs dfs -ls /sales
2026-08-02 14:43:54,859 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using bui
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ hdfs dfs -cat /sales/sales_data.txt
2026-08-02 14:44:03,408 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using bui
cat: '/sales/sales_data.txt': No such file or directory
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ jps
5204 NameNode
5513 SecondaryNameNode
7147 Jps
prisha-kalola@prisha-kalola-VMware-Virtual-Platform: ~/BDA Practicals/Practical_2$ ls -R ~/hadoopdata
/home/prisha-kalola/hadoopdata:
hdfs

/home/prisha-kalola/hadoopdata/hdfs:
datanode namenode

/home/prisha-kalola/hadoopdata/hdfs/datanode:
current

/home/prisha-kalola/hadoopdata/hdfs/datanode/current:
BP-270177611-127.0.1.1-1784043942624 VERSION

/home/prisha-kalola/hadoopdata/hdfs/datanode/current/BP-270177611-127.0.1.1-1784043942624:
current scanner.cursor tmp

/home/prisha-kalola/hadoopdata/hdfs/datanode/current/BP-270177611-127.0.1.1-1784043942624/current:
VERSION finalized rbw

```

```

prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ ls /usr/local/hadoop/logs
SecurityAuth-prisha-kalola.audit
hadoop-prisha-kalola-datanode-prisha-kalola-VMware-Virtual-Platform.log
hadoop-prisha-kalola-datanode-prisha-kalola-VMware-Virtual-Platform.out
hadoop-prisha-kalola-datanode-prisha-kalola-VMware-Virtual-Platform.out.1
hadoop-prisha-kalola-namenode-prisha-kalola-VMware-Virtual-Platform.log
hadoop-prisha-kalola-namenode-prisha-kalola-VMware-Virtual-Platform.out
hadoop-prisha-kalola-namenode-prisha-kalola-VMware-Virtual-Platform.out.1
hadoop-prisha-kalola-nodenanager-prisha-kalola-VMware-Virtual-Platform.log
hadoop-prisha-kalola-nodenanager-prisha-kalola-VMware-Virtual-Platform.out
hadoop-prisha-kalola-nodenanager-prisha-kalola-VMware-Virtual-Platform.out.1
hadoop-prisha-kalola-resourcenanager-prisha-kalola-VMware-Virtual-Platform.log
hadoop-prisha-kalola-resourcenanager-prisha-kalola-VMware-Virtual-Platform.out
hadoop-prisha-kalola-resourcenanager-prisha-kalola-VMware-Virtual-Platform.out.1
hadoop-prisha-kalola-resourcenanager-prisha-kalola-VMware-Virtual-Platform.out.2
hadoop-prisha-kalola-secondarynamenode-prisha-kalola-VMware-Virtual-Platform.log
hadoop-prisha-kalola-secondarynamenode-prisha-kalola-VMware-Virtual-Platform.out
hadoop-prisha-kalola-secondarynamenode-prisha-kalola-VMware-Virtual-Platform.out.1
userlogs
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ tail -50 /usr/local/hadoop/logs/hadoop-pris
ware-Virtual-Platform.log
2026-08-02 14:40:58,180 INFO org.eclipse.jetty.server.handler.ContextHandler: Started o.e.j.s.ServletContextHandler@2449c7f7{
doop/share/hadoop/hdfs/webapps/static/,AVAILABLE}
2026-08-02 14:40:58,995 INFO org.eclipse.jetty.server.handler.ContextHandler: Started o.e.j.w.WebAppContext@7fcff1b9{datanode
adoop/hdfs/webapps/datanode/,AVAILABLE}{file:/usr/local/hadoop/share/hadoop/hdfs/webapps/datanode}
2026-08-02 14:40:59,039 INFO org.eclipse.jetty.server.AbstractConnector: Started ServerConnector@58a55449{HTTP/1.1, (http/1.1
2026-08-02 14:40:59,039 INFO org.eclipse.jetty.server.Server: Started @12180ms
2026-08-02 14:40:59,012 WARN org.apache.hadoop.hdfs.server.datanode.web.DatanodeHttpServer: Got null for restCsrfPreventionFi

prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ stop-dfs.sh
Stopping namenodes on [localhost]
Stopping datanodes
jpsStopping secondary namenodes [prisha-kalola-VMware-Virtual-Platform]
2026-08-02 14:58:16,172 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-ja
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ jps
7827 Jps
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ rm -rf ~/hadoopdata/hdfs/namenode/*
rm -rf ~/hadoopdata/hdfs/datanode/*
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ ls -R ~/hadoopdata/hdfs
/home/prisha-kalola/hadoopdata/hdfs:
datanode namenode

/home/prisha-kalola/hadoopdata/hdfs/datanode:

/home/prisha-kalola/hadoopdata/hdfs/namenode:
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ hdfs namenode -format
2026-08-02 14:51:04,511 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG: host = prisha-kalola-VMware-Virtual-Platform/127.0.1.1
STARTUP_MSG: args = [-format]
STARTUP_MSG: version = 3.4.1
STARTUP_MSG: classpath = /usr/local/hadoop/etc/hadoop:/usr/local/hadoop/share/hadoop/common/lib/kerby-pkix-2.0.3.jar:/usr/l
/jul-to-slf4j-1.7.36.jar:/usr/local/hadoop/share/hadoop/common/lib/failureaccess-1.0.jar:/usr/local/hadoop/share/hadoop/connor
:/usr/local/hadoop/share/hadoop/common/lib/hadoop-auth-3.4.1.jar:/usr/local/hadoop/share/hadoop/common/lib/jakarta.activation

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prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ jps
8563 Jps
8355 SecondaryNameNode
8187 DataNode
8876 NameNode
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ cd ~/home/prisha-kalola/BDA Practicals/Practical_2
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ hdfs dfs -mkdir /sales
2026-08-02 14:53:36,057 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ hdfs dfs -ls /
2026-08-02 14:53:44,037 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java
Found 1 items
drwxr-xr-x    - prisha-kalola supergroup          0 2026-08-02 14:53 /sales
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ hdfs dfs -put sales_data.txt /sales
2026-08-02 14:53:53,198 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ hdfs dfs -ls /sales
2026-08-02 14:54:07,475 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java
Found 1 items
-rw-r--r--    1 prisha-kalola supergroup          63 2026-08-02 14:53 /sales/sales_data.txt
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ hdfs dfs -cat /sales/sales_data.txt
2026-08-02 14:54:17,618 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java
ProductA 120
ProductB 80
ProductC 150
ProductA 280
ProductB 90
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ hdfs dfs -cp /sales/sales_data.txt /sales/s
2026-08-02 14:54:30,413 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ hdfs dfsadmin -report
2026-08-02 14:56:54,315 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java
Configured Capacity: 20954300416 (19.52 GB)
Present Capacity: 3217981440 (3.00 GB)
DFS Remaining: 3217956864 (3.00 GB)
DFS Used: 24576 (24 KB)
DFS Used%: 0.00%
Replicated Blocks:
    Under replicated blocks: 0
    Blocks with corrupt replicas: 0
    Missing blocks: 0
    Missing blocks (with replication factor 1): 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
Erasure Coded Block Groups:
    Low redundancy block groups: 0
    Block groups with corrupt internal blocks: 0
    Missing block groups: 0
    Low redundancy blocks with highest priority to recover: 0
    Pending deletion blocks: 0
-----
Live datanodes (1):

Name: 127.0.0.1:9866 (localhost)
Hostname: prisha-kalola-VMware-Virtual-Platform
Decommission Status : Normal

```



```
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Sun Aug 02 14:57:15 IST 2026
Last Block Report: Sun Aug 02 14:52:25 IST 2026
Nun of Blocks: 0

prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ jps
8355 SecondaryNameNode
9657 Jps
8187 DataNode
8076 NameNode
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ hdfs dfs -df -h
2026-08-02 14:57:33,472 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-j
Filesystem                Size      Used Available Use%
hdfs://localhost:9808    19.5 G    24 K      3.0 G    0%
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ stop-dfs.sh
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [prisha-kalola-VMware-Virtual-Platform]
2026-08-02 14:58:02,253 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-j
prisha-kalola@prisha-kalola-VMware-Virtual-Platform:~/BDA Practicals/Practical_2$ jps
10191 Jps
```

## Key Questions / Analysis

### Q1. Why is Hadoop preferred for Big Data storage?

Hadoop can store and process very large amounts of data across multiple computers. It is fast, reliable, and cost-effective.

---

### Q2. What is the role of NameNode and DataNode in HDFS?

The NameNode manages file information and keeps track of where data is stored. The DataNode stores the actual data blocks and sends them when requested.

---

### Q3. What is the difference between HDFS and traditional file systems?

HDFS stores data across multiple machines, while a traditional file system stores data on a single computer. HDFS is designed for handle.

---



**Q4. What are the advantages of distributed storage?**

Distributed storage improves speed, reliability, scalability, and allows data to remain available even if one machine fails.

---

**Supplementary Questions****1. Configure Hadoop pseudo-distributed mode.**

Pseudo-distributed mode allows Hadoop services to run on a single computer for learning, testing, and development.

---

**2. Change HDFS replication factor and observe results.**

Changing the replication factor changes the number of copies of each data block stored in HDFS, improving data availability and fault tolerance.

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**3. Upload a large dataset into HDFS.**

A large dataset is uploaded into HDFS using Hadoop commands so it can be stored and processed efficiently.

---

**4. Analyze HDFS storage utilization.**

Storage utilization is checked to understand how much HDFS space is used and how much is available.

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**Conclusion**

In this practical, Hadoop was successfully installed and configured in a Single-Node Cluster environment. HDFS operations such as creating directories, uploading, downloading, moving, renaming, and deleting files were performed successfully. Hadoop services and storage utilization were also monitored. This practical helped in understanding how Hadoop manages and stores large amounts of data efficiently using

HDFS.

## **Key Skills to be Addressed**

- Hadoop Installation and Configuration
- Environment Variable Configuration
- HDFS Administration
- Distributed Storage Management
- Hadoop Cluster Monitoring

## **Dataset / TestData**

**sales\_data.txt**

**ProductA 120**

**ProductB 80**

**ProductC 150**

**ProductA 200**

**ProductB 90**

**Format**

**TXT/CSV Files (mb/gb)**